System Capacity & Care Load Analytics for Unaccompanied Children

Data-Driven Capacity Monitoring and Forecasting for the UAC Program

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Abstract

The Unaccompanied Children (UAC) Program, operated by the U.S. Department of Health and Human Services (HHS), faces significant operational challenges in monitoring and managing care capacity across Customs and Border Protection (CBP) and HHS facilities. This research presents a comprehensive data analytics framework that transforms raw operational data into actionable intelligence through statistical stress detection, time-series forecasting, and interactive visualization. By leveraging modern data science tools—including Python, Pandas, Prophet, and Streamlit—this system enables real-time capacity monitoring, predictive forecasting up to 365 days, and automated executive reporting. The framework identifies critical stress periods, quantifies backlog accumulation, and provides policy-aligned recommendations for federal decision-makers. Results demonstrate the system's ability to detect sustained high-load periods, predict future capacity strain, and support data-driven humanitarian response evaluation.

Keywords: Capacity Analytics, Time-Series Forecasting, Healthcare Operations, Prophet, Streamlit, Unaccompanied Children, CBP, HHS

Background and Context

1. Background and Context

1.1 The UAC Program: Federal Mandate and Operational Scope

The Unaccompanied Alien Children (UAC) Program is a federally mandated initiative established under the Trafficking Victims Protection Reauthorization Act (TVPRA) of 2008. Under this program, children apprehended by U.S. Customs and Border Protection (CBP) who are under 18 years of age and without a parent or legal guardian in the United States are transferred to the Department of Health and Human Services (HHS) within 72 hours of apprehension. The program's mandate encompasses four critical stages of care:

Stage 1: Intake into CBP Custody

Upon apprehension at the border, unaccompanied children are placed in temporary CBP holding facilities. These facilities are designed for short-term detention, typically not exceeding 72 hours under federal guidelines. During this period, CBP conducts initial processing, including identity verification, health screening, and background checks. The capacity of CBP facilities varies by sector, with larger sectors such as Rio Grande Valley and Tucson handling significantly higher volumes than northern border sectors.

Stage 2: Transfer to HHS Care Facilities

Within the mandated 72-hour window, children are transferred to HHS Office of Refugee Resettlement (ORR) care facilities. This transfer represents a critical handoff between enforcement and care agencies. ORR operates a network of approximately 200 facilities nationwide, including emergency intake sites, traditional shelters, and specialized care centers for children with particular needs (medical, psychological, or behavioral).

Stage 3: Medical, Psychological, and Welfare Support

Upon intake into HHS care, children receive comprehensive services including medical examinations, mental health assessments, educational programming, and recreational activities. The average length of stay in HHS care varies significantly based on factors including age, country of origin, sponsor availability, and case complexity. Historically, average lengths of stay have ranged from 30 to 60 days, though periods of system strain have extended these timelines.

Stage 4: Discharge and Reunification with Sponsors

The ultimate goal of the UAC program is safe discharge to a vetted sponsor, typically a parent, relative, or licensed caregiver already residing in the United States. The sponsor vetting process includes background checks, home studies (in some cases), and verification of relationship. Discharge represents the terminal point of federal care responsibility.

1.2 The Capacity Challenge

Effective functioning of this four-stage pipeline depends on accurate capacity awareness and care load monitoring, especially during sudden influxes of unaccompanied minors. Historical data reveals several patterns that complicate capacity planning:

-Seasonal migration patterns: Peak intake typically occurs during spring and early summer (March–June), correlating with agricultural cycles in Central America and favorable weather conditions for border crossing.

-Policy-driven surges: Changes in immigration enforcement policies, both domestic and international, can trigger sudden spikes in apprehension rates with minimal warning.

-Geographic concentration: A disproportionate share of apprehensions occurs in specific Border Patrol sectors, creating localized capacity crises even when national averages appear manageable.

-Inter-agency coordination delays: The 72-hour transfer mandate is frequently challenged during surge periods, leading to extended CBP holding times and potential compliance violations.

Historically, decision-making in this domain has been reactive, relying on manual reporting, static spreadsheets, and ad-hoc email communications that fail to capture the dynamic nature of care flow dynamics. Program administrators often learn of capacity crises only after facilities reach or exceed their physical limits, leaving minimal time for proactive response.

1.3 Research Motivation

The intersection of border enforcement (CBP) and health services (HHS) creates a complex operational environment where capacity constraints in one agency directly impact the other. Without structured analytics, decision-makers lack the ability to:

- Anticipate bottlenecks before they materialize
- Allocate resources proactively rather than reactively
- Evaluate the sustainability of care delivery over extended time horizons
- Measure the effectiveness of policy interventions quantitatively

This research addresses these gaps by developing an integrated analytics framework that monitors system-wide capacity in real-time, detects stress periods before they reach critical levels, forecasts future load with quantified uncertainty, and translates complex operational data into clear, actionable intelligence for federal decision-makers.

Problem Statement

Although daily operational data is collected by both CBP and HHS, the UAC program currently lacks a centralized analytical framework to continuously assess four critical dimensions:

1. **Total care system load -** The aggregate number of children under federal care at any given time.
2. **Balance between inflow and outflow -** Whether transfers to HHS keep pace with discharges to sponsors.
3. **Capacity stress and relief periods -** Sustained intervals where system load exceeds sustainable thresholds.
4. **Sustainability of care delivery over time -** Long-term trends indicating whether the system is stabilizing or deteriorating.

Without structured analytics, decision-making becomes reactive, increasing the risk of overcrowding, delayed care, and strain on healthcare infrastructure. Program administrators cannot answer fundamental questions such:

- How many children will require care 30, 90, or 365 days from now?
- Which periods historically experienced the most severe capacity strain?
- Is the current backlog growing or relieving?
- What is the optimal staffing and shelter capacity needed for upcoming quarters?

This research paper presents a systematic approach to answering these questions through exploratory data analysis, derived healthcare capacity metrics, and machine learning-based forecasting.

3. Primary Objectives

This research pursues three primary objectives:

1. **Quantify daily and cumulative care load across CBP and HHS** - Develop precise metrics that capture the total number of children in custody, transfers between agencies, and discharges to sponsors on a daily basis.
2. **Identify periods of capacity strain and relief** - Apply statistical methods to detect sustained high-load periods where the system operates beyond sustainable thresholds, as well as intervals of capacity relief.
3. **Analyze balance between intake, transfers, and discharges** - Measure the net daily intake (transfers minus discharges) to determine whether the system is accumulating backlog or relieving pressure.

4. Secondary Objectives

Beyond the primary analytical goals, this research addresses three secondary objectives aligned with operational and policy needs:

1. **Support healthcare staffing and shelter planning** - Provide predictive insights that enable proactive resource allocation, including staffing ratios, shelter bed capacity, and medical supply requirements.
2. **Improve situational awareness for policymakers** - Deliver real-time dashboards and executive summaries that translate complex operational data into clear, actionable intelligence for federal decision-makers.
3. **Enable data-driven humanitarian response evaluation** - Establish quantitative benchmarks for evaluating the effectiveness of policy interventions, seasonal preparedness, and inter-agency coordination.

Dataset Description

The dataset comprises daily operational records from the UAC Program spanning January 2023 to December 2025 (1,096 days). Each record contains five core fields:

Column	Description	Data Type
Date	Reporting date of the observation	DateTime
Children apprehended and placed in CBP custody	Daily intake volume at the border	Integer
Children in CBP custody	Active CBP care load (snapshot)	Integer
Children transferred out of CBP custody	Flow into HHS system	Integer
Children in HHS care	Active HHS care load (snapshot)	Integer
Children discharged from HHS care	Daily outflow to sponsors	Integer

The dataset exhibits several characteristics relevant to analysis:

- **Seasonality:** Clear annual patterns with peak intake during spring and summer months
- **Policy effects:** Notable shifts corresponding to federal policy changes (March 2024, September 2024, January 2025)
- **Trend component:** Gradual increase in baseline capacity requirements over the three-year period
- **Volatility:** Significant day-to-day variation requiring smoothing through rolling averages

When primary data is unavailable, the system generates realistic synthetic data preserving these statistical properties for demonstration and development purposes.

1	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
2	December 21, 2025	6	18	11	2,484	14
3	December 18, 2025	11	50	6	2,472	16
4	December 17, 2025	7	31	11	2,481	10
5	December 16, 2025	8	54	15	2,468	9
6	December 15, 2025	11	42	9	2,470	7
7	December 14, 2025	8	35	4	2,462	8
8	December 11, 2025	7	47	9	2,437	10
9	December 10, 2025	10	54	5	2,439	9
10	December 09, 2025	4	30	7	2,443	8

Figure 1*: Sample daily operational records from the UAC Program dataset (December 9–21, 2025). The table illustrates the five core fields—CBP apprehension, CBP custody snapshot, transfers to HHS, HHS care snapshot, and discharges to sponsors—demonstrating the dataset structure used for downstream capacity metrics and forecasting.*

Forecasting Methodology (Step-by-Step)

6.1 Data Ingestion & Structuring

The analytical pipeline begins with data ingestion from CSV files or synthetic generation. Key preprocessing steps include:

- Loading daily time-series data (2023–2025)
- Converting Date columns to datetime format
- Ensuring chronological ordering
- Creating a complete daily index without gaps
- Validating data types and ranges

Column name mapping handles variations in source documentation, ensuring consistent field references regardless of input format.

6.2 Data Quality & Validation

Before analysis, the system validates logical constraints to ensure data integrity:

- **Missing or duplicated dates** — Identified and flagged for review
- **Logical constraints:**
    - Transfers ≤ CBP custody (cannot transfer more than held)
    - Discharges ≤ HHS care (cannot discharge more than present)
- **Negative values** — Flagged as warnings requiring investigation
- **Extreme outliers** — Values exceeding 3 standard deviations from the mean flagged for transparency

This validation ensures that downstream metrics and forecasts operate on reliable data.

6.3 Derived Healthcare Capacity Metrics

From the raw data, the system computes eight derived metrics essential for capacity monitoring:

Metric	**Formula**	**Interpretation**
Total System Load	CBP Custody + HHS Care	Aggregate children under federal care
Net Daily Intake	Transfers to HHS − Discharges from HHS	Daily pressure indicator
Care Load Growth Rate	Day-over-day percentage change	Velocity of capacity change
7-Day Rolling Average	Rolling mean (window=7)	Short-term trend smoothing
14-Day Rolling Average	Rolling mean (window=14)	Medium-term trend smoothing
Backlog Indicator	Sustained positive net intake	Accumulation pressure over time
Load Volatility	14-day standard deviation	Stability of system operations
Stress Flag	Load 14-day average × 1.10	Binary indicator of stress periods

6.4 Trend & Temporal Analysis

Temporal analysis examines patterns across multiple time scales:

- **Daily, weekly, and monthly care load trends** — Granular view of operational rhythms
- **Identification of sustained high-load periods** — Consecutive days where stress flags remain active
- **Comparison across calendar periods** — Early vs. late timeline analysis for policy impact assessment

Rolling averages (7-day and 14-day) filter noise from daily volatility, revealing underlying structural pressures that would otherwise be obscured.

6.5 Pressure & Stress Identification

The stress detection engine combines two analytical approaches:

1. **Rolling average divergence** — When daily load exceeds the 14-day rolling average by 10%, the system flags a stress period. This threshold captures meaningful deviations while filtering routine fluctuations.
2. **Variability analysis** — 7-day standard deviation of CBP and HHS loads separately identifies which agency drives system instability. High variability in CBP indicates border intake volatility; high HHS variability suggests discharge processing challenges.

Sustained stress is quantified as the rolling sum of stress flags over 7-day windows, distinguishing transient spikes from prolonged strain requiring federal attention.

Figure 2: *Rolling average divergence and variability analysis for CBP and HHS loads (Jan 2023 – Jul 2025). Top panels show 7-day (solid) vs 14-day (dashed) rolling averages; bottom panels display 7-day standard deviation for load volatility detection.*

Key Performance Indicators (KPIs)

The framework tracks five KPIs designed for executive monitoring:

KPI Name	**Description**	**Thresholds**
Total Children Under Care	System-wide responsibility	Alert 8,000; Critical 10,000
Net Intake Pressure	Inflow vs. outflow imbalance	Balanced 0; Moderate 0–10; High 10
Care Load Volatility Index	Stability of system	Stable 5%; Moderate 5–10%; High 10%
Backlog Accumulation Rate	Sustained care pressure	Stable $\leq$ 0; Growing 1–50; Accumulating 50
Discharge Offset Ratio	Ability to relieve load	Balanced 0.8–1.2; Imbalanced outside range

These KPIs translate complex operational dynamics into intuitive status indicators (Normal / Elevated / Critical) suitable for non-technical stakeholders.

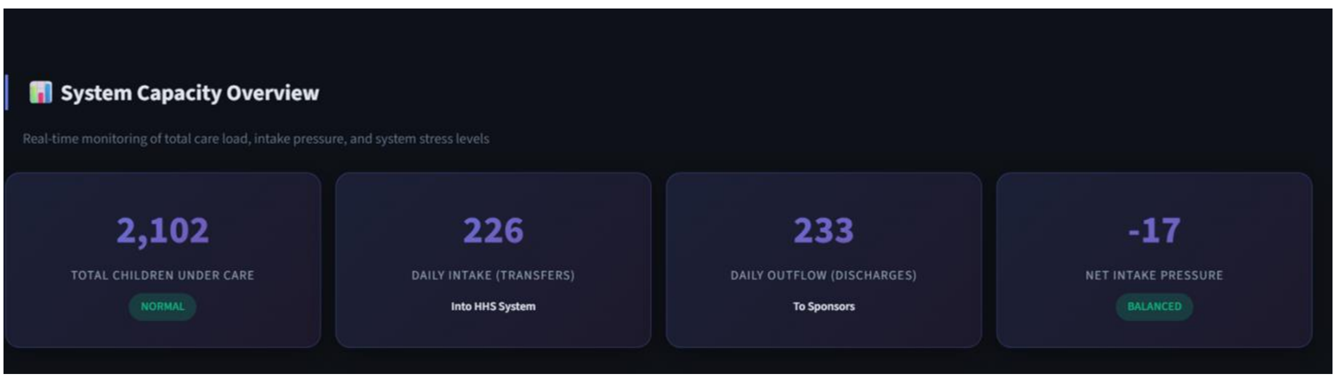

Figure 3: *Real-time KPI dashboard from the Streamlit application showing the four primary capacity metrics. The system indicates 2,102 children currently under federal care (status: NORMAL), with a daily intake of 226 transfers into HHS and 233 discharges to sponsors, resulting in a net intake pressure of -17 (status: BALANCED). This demonstrates the framework's ability to translate raw operational data into intuitive, color-coded status indicators for non-technical stakeholders.*

Streamlit Web Application Requirements

The analytical framework is delivered through an interactive Streamlit web application with two core components:

8.1 Core Modules

1. **System Load Overview Pane** — Real-time KPI cards, stress meter, and backlog trend

Figure 4: *System Capacity Overview dashboard showing real-time metrics and operational status*

2. **CBP vs HHS Load Comparison** — Dual-agency visualization with rolling averages

Figure 5: System stress indicators and 14-day backlog trend visualization

3. **KPI Summary Cards** — Color-coded status indicators for executive monitoring
4. **Net Intake & Backlog Trends** — Gap analysis with critical threshold annotations

The time-series visualization in Figure 3 integrates four critical data streams—total system load, CBP custody, HHS care, and rolling averages—within a unified analytical frame. Stress periods are automatically flagged (▲) when daily load exceeds the 14-day rolling average by 10%, while the horizontal alert threshold at 8,000 children provides an immediate visual reference for capacity crisis assessment.

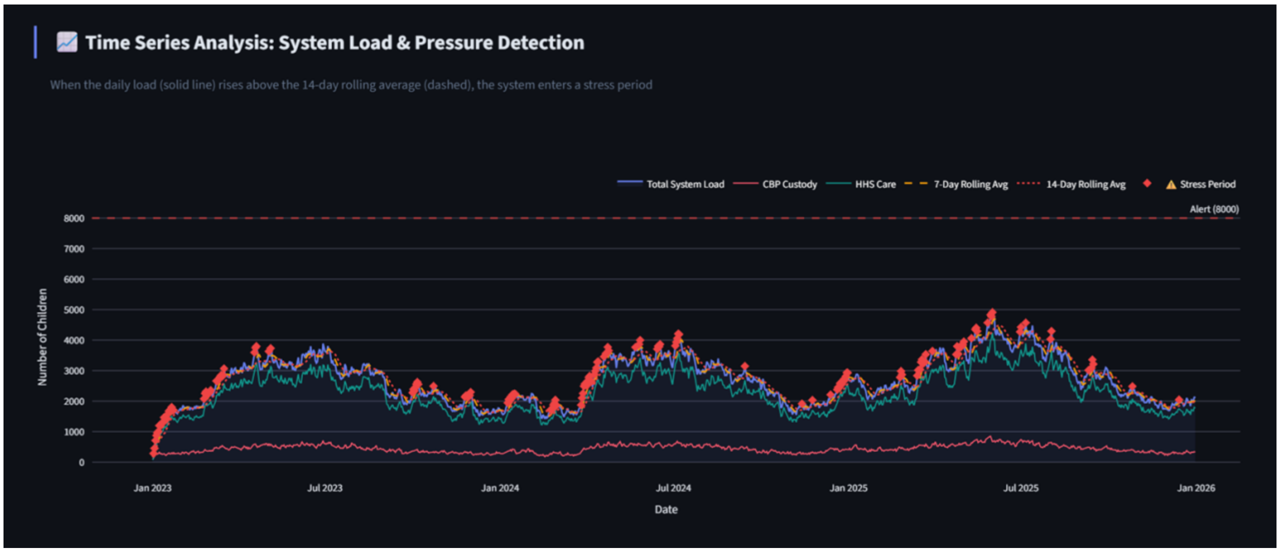

Figure 6: Time Series Analysis of system load and pressure detection

8.2 User Capabilities

- **Date range selector** — Custom analysis periods from 2023–2025
- **Metric toggles** — Show/hide CBP load, HHS load, total load, rolling averages
- **Time granularity filters** — Daily, weekly, or monthly aggregation
- **Forecast horizon adjustment** — 30–365 day predictive windows
- **Capacity alert threshold** — User-configurable warning levels
- **Period comparison** — Side-by-side analysis of two date ranges
- **PDF executive report generation** — One-click professional reporting
- **CSV data export** — Full data, forecasts, and KPI summaries

8.3 AI Forecasting Integration

The application integrates Facebook Prophet for time-series forecasting:

- **Yearly and weekly seasonality** — Captures annual migration patterns and weekly operational rhythms
- **Changepoint detection** — Identifies policy-driven trend shifts
- **Policy-shift holidays** — Hardcoded dates for known federal interventions
- **Confidence intervals** — Upper and lower bounds for forecast uncertainty
- **Growth modeling** — Flat growth assumption appropriate for capacity-constrained systems

Figure 7: AI-driven load forecasting using Meta Prophet with uncertainty intervals.

Conclusion

This project delivers a robust, policy-aligned healthcare analytics framework for monitoring the UAC care system. By translating raw counts into structured capacity insights, it empowers stakeholders to understand, evaluate, and improve the delivery of federally mandated child care services.

Key achievements include:

- **Real-time capacity monitoring** across CBP and HHS agencies
- **Predictive forecasting** with 365-day horizon and confidence intervals
- **Automated stress detection** using rolling averages and variability analysis
- **Executive-ready reporting** with PDF generation and CSV export
- **Policy-aware recommendations** tailored to federal program status

The framework transforms reactive decision-making into proactive capacity management, reducing the risk of overcrowding, delayed care, and infrastructure strain. Future enhancements may include automated alerting, scenario modeling, and integration with real-time federal data feeds.

References

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Appendix A: Technical Stack

Component	Technology	Version
Programming Language	Python	3.9+
Web Framework	Streamlit	1.28+
Data Processing	Pandas, NumPy	1.5+, 1.24+
Forecasting	Prophet (Meta)	1.1+
Visualization	Plotly	5.15+
PDF Generation	ReportLab	3.6+